EPSR-test-ex34 fmole: prompts, repulsive core, missing template

Program(s): fmole | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises further `fmole` paths [1]: the file name and number of shakes typed on stdin, the intramolecular repulsive core potential switched on through the `ecore dcore` line of the `.ato` file, and a molecule whose template file is missing. This complements the standard route of ex33 for the geometry relaxation used before EPSR runs [2].

Code exercised

Fortran (original, fortran_src/):

- `fmole.f`: `fmole` (stdin prompts), `initpt_fmole`, `update_ato` (template cannot be opened: bonds kept)
- `ato_manipulation_rev4.f`: `generate_mp` (repulsive core from `ecore`, `dcore`), `fmolec1`, `adjust_intra_wts`
- `molfit.f`: `molfit`; `file_and_directory_commands.f`: `search_for_files` (file prompt / directory listing), `get_full_filename`
- `ato_read_and_save.f`: `minitc` (read `.ato`), `moloutc` (write `.ato`)

Rust (port, rust/epsr-core/src/):

- `tools/fmole.rs`: `run`, `initpt_fmole`, `update_ato`
- `potential.rs`: `generate_mp`; `energy.rs`: `fmolec1`, `adjust_intra_wts`; `montecarlo.rs`: `molfit`
- `tools/setup_common.rs`: `search_for_files`, `get_full_filename`, `command_text`, `read_text`; `ato.rs`: `minitc`, `moloutc`, `ftype`

Physical system

Single molecules of the EPSR-test system [3] at 300 K: one dimethyl sulphoxide molecule (10 atoms, box 21.544 Å) and one water molecule (3 atoms, box 14.422 Å). In `DMSOcore.ato` the `ecore dcore` line is set to 0.3 2.5 (instead of 0 0), so the repulsive Weeks-Chandler-Andersen-type core of `generate_mp` [4] acts on the non-bonded H...H pairs in addition to the harmonic bond and angle restraints.

Inputs

`DMSO.ato` from EPSR-test; `DMSOcore.ato` = `DMSO.ato` with `ecore dcore` = `0.3 2.5` (the only changed line); `Water.ato` = `WaterMolecule.ato` renamed, so its template is absent. Commands: `fmole` < `fmole1.stdin` (file `DMSOcore` typed — no listing — and 150 shakes asked on stdin) and `fmole Water` < `fmole2.stdin` (30 shakes); `update_ato` cannot open the template and keeps the bonds of the file.

How the accuracy was verified

`commands.txt` (2 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Two program calls in one pass; compared outputs: the rewritten `DMSOcore.ato` and `Water.ato` and the empty `dirlist.txt`, identical including coordinates and the random-number state.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).

[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).

[4] J. D. Weeks, D. Chandler and H. C. Andersen, "Role of repulsive forces in determining the equilibrium structure of simple liquids", J. Chem. Phys. 54, 5237-5247 (1971).

[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.